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Paleomagnetism of some Grenville Province Rocks

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Anorthositic rocks 20 to 30 km south of the Grenville Front in the Grenville Province of Ontario have a stable remanent magnetization which yields a paleomagnetic pole at 161° E, 8° N. This pole is close to other poles from the Precambrian of North America of 1050 m.y. age. Paleomagnetic results have also been obtained from two rock units in the southern part of the Grenville Province of Ontario. These are a pyroxenite exposed near Wilberforce and the Tudor Gabbro exposed near Madoc. The directions of remanent magnetization yield pole positions as follows: Wilberforce pyroxenite—148° E, 14.5° S; Tudor Gabbro—137° E, 17° N. These pole positions, together with others from the literature, which have been derived from rock units 200 km or more south of the Grenville Front, plot farther west in the Pacific than do poles of inferred similar age from other regions of North America. A possible interpretation of these results is that the southern part of the Grenville province was formed some 30° SE of its present position.

Les roches anorthositiques situées entre 20 km et 30 km au Sud du Front de Grenville, dans la Province de Grenville d'Ontario, ont une magnétisation rémanente stable donnant un pôle magnétique à 161° E, 8° N. Cette attitude est en accord avec celle des autres pôles pré-cambriens d'Amérique du Nord de 1050 millions d'années d'âge. D'autres résultats paléomagnétiques furent obtenus à partir de 2 unités lithologiques du Sud de la Province de Grenville d'Ontario, soit une pyroxénite affleurant près de Wilberforce et le Gabbro de Tudor affleurant près de Madoc. Les directions de magnétisation rémanente donnent les attitudes de pôles magnétiques suivantes: 148° E, 14½° S pour la pyroxénite de Wilberforce et 137° E, 17° S pour le Gabbro de Tudor. Les positions de ces pôles, et de ceux, déjà publiés, d'unités lithologiques située à 200 km et plus au Sud du Front de Grenville, se reportent plus à l'Ouest dans le Pacifique que ceux des autres régions d'Amérique du Nord de même âge. Une façon d'interpréter ces résultats serait de supposer que le Sud de la Province de Grenville fut formé à 30 SE de sa position actuelle.

[Traduit par le journal]

Introduction

The plate tectonic model is difficult to contest as applied to earth history for the last 100 m.y. or so as it is the only model which satisfactorily explains the marine magnetic anomalies and the present pattern of global seismicity. In addition it satisfies the classical arguments for continental drift. The question as to whether there have been pre-Mesozoic episodes of drift is a more open one and much less evidence can be mustered to test the hypothesis. Land based paleomagnetic measurements provide one line of evidence which may be used to test plate–plate interactions for these more remote portions of geological time. For paleomagnetic data to be effective in this regard, the spreading pole must not have coincided with the paleomagnetic pole and the displacements must have been comparable in magnitude to those that have occurred from Permian time to the present.

The form of the paleomagnetic wandering path with respect to North America (exclusive of the Grenville Province) from the end of Precambrian time to about 1.5 b.y. ago is moderately well established (Spall 1971, Robertson and Fahrig 1971). Within the Grenville Province rocks of Early Precambrian age

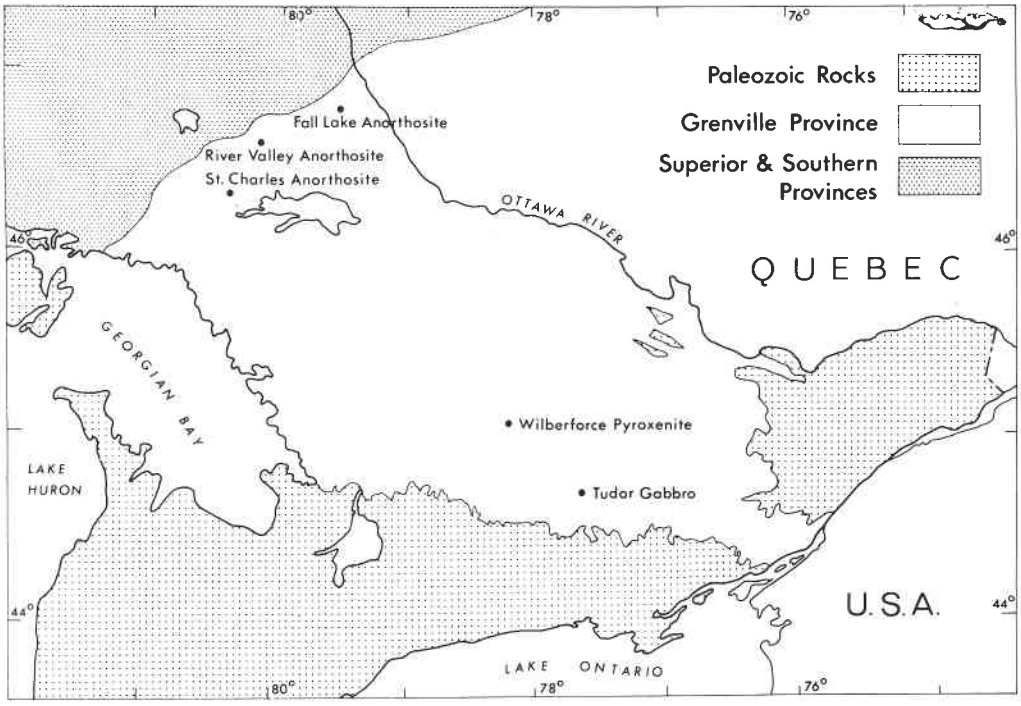


FIG. 1. Locality map showing the locations of the five rock units for which paleomagnetic results are reported on in this paper.

have not been found more than a few miles south of the Grenville Front. Apart from these rocks, the oldest radiometric ages obtained from igneous rocks in the Grenville Province do not exceed 1.8 b.y. (Krogh and Davis 1971) and the province as a whole was effectively sealed to argon loss about 950 m.y. ago (Emslie 1970). Thus, rocks of the Grenville Province should have acquired their remanent magnetization during that period of time for which the field configuration with respect to North America is moderately well known and, significantly for the problem at hand, the configuration varied rather markedly during that time interval. Anomalous (with respect to North America) pole positions calculated from the remanence directions in rocks from the southern part of the Grenville Province have led Irving *et al.* (1972) and Palmer and Carmichael (1972) to postulate that part of the Grenville Province was once displaced relative to the remainder of the Canadian shield. In this paper we present the data which led us to that conclusion. Because lithological types northwest of the Grenville Front can be traced for some distance into the Grenville Province (100 km in the case of the distinctive iron formations of the Labrador trough) it is clear that the Grenville Front itself does not represent the suture between the two ancient plates. Accordingly recent sampling was undertaken in the northern part of the Grenville Province in rocks which are intruded into probable Huronian equivalents and which should yield pole positions on the North American curve. These 'non-anomalous' results are also reported in this paper.

Rock units from which paleomagnetic results have been obtained by us are from the localities indicated in Fig. 1. The two rock units from the southern portion of the Grenville Province of Ontario are a pyroxenite exposed near Wilberforce, Ontario and the Tudor Gabbro which crops out in portions of Tudor and Lake Townships. The three northern localities are in anorthosite to gabbroic anortho-

site bodies mapped by Lumbers (1971*a*, *b*, *c*, 1972): the St. Charles anorthosite, the River Valley anorthosite, and the Fall Lake Complex.

Southern Grenville Province—Wilberforce Pyroxenite

Attention was first drawn to this area upon publication of the federal government aeromagnetic map covering the Wilberforce area (Map 110G, Aeromagnetic series, Geological Survey of Canada, 1953). An area to the south of Allen Lake, centered on coordinates 45°7′ N, 78°15′ W, was characterized by a sharp negative (1800 γ below background) magnetic anomaly. Because the rich titanium deposits of the Allard Lake area of Quebec were also characterized by negative magnetic anomalies, a considerable amount of exploratory drilling was undertaken in the area in the early 1950's. No massive concentrations of Fe–Ti oxides were found.

Mason (1959) demonstrated from a combination of ground magnetic and electromagnetic surveys and surface sampling that the anomalies could be accounted for by the surface rock, none of which contained high concentrations of Fe–Ti oxides. Work by Mason (1959) and Irwin (1963) demonstrated that the intense remanence in the surface rock was carried by fine lamellae and rods exsolved within the pyroxenes and to a lesser extent in the amphiboles. In thin section the rocks are seen to consist of a granoblastic aggregate of clinopyroxene, amphibole, and chlorite. Minor amounts of olivine are present in some sections examined. The work of Mason (1959) and Irwin (1963) included the collection of 14 oriented block samples. These samples were taken from outcrops spread over a distance of approximately 1 km with no 2 samples closer than 30 m apart. The azimuthal component of orientation was determined with reference to a baseline or by a Brunton compass held at a distance from the outcrop. In the laboratory 3 to 5 specimens were drilled from each sample and the NRM directions were measured on an astatic magnetometer. All samples, when initially measured, exhibited reversed directions of magnetization but were rather poorly grouped. Alternating field demagnetization was performed by Irwin (1963) using a demagnetizing unit which incorporated a 2-axis tumbling unit; field reduction was accomplished by physically withdrawing the energizing coil by a system of pulleys. Directions of remanent magnetization were determined in these specimens after three stages of demagnetization carried out at peak fields of 250, 600, and 960 Oe. Sample average directions calculated after the 250 Oe stage of demagnetization are shown in Fig. 2. Results obtained after the 600 Oe stage showed a slight reduction in precision over the results obtained after the lower field treatment. Greater scatter still was observed after the 960 Oe treatment.

In 1971, virgin specimens were drilled from 4 of the 14 original samples which still had legible orientation markings. The directions of NRM were measured using a PAR Model SM-1 spinner magnetometer and the a.f. demagnetization was accomplished using a unit incorporating a 3-axis tumbling system, larger coils for cancellation of the earth's field, and a motor driven inductrol for raising and lowering the alternating field. The demagnetization of these specimens was carried out at the following peak fields: 50, 100, 200, 300, 400, 500, 600, 800, 1000, and 1200 Oe. Directions and intensities of remanent magnetization were measured after each step. Convergence to the mean direction illustrated in Fig. 2 was achieved after the 50 Oe or the 100 Oe treatment and the directions did not significantly change until after the 800 Oe treatment when scattering began in 2 of the 4 samples. In the remaining 2 samples, scattering set in after the 1000 Oe treatment. The conclusion we draw is that the mean direction obtained using the 250 Oe data obtained on the older and rather imperfect equipment is a stable end point.

However, it was desirable to check the accuracy of this result by improved field orienting procedures. Accordingly, in the summer of 1972 the area was revisited and 4 or 5 field-drilled cores (samples) were obtained from each of three sites. The azimuthal component of orientation of these field-drilled cores was determined by sun compass. Generally each core provided two specimens in which the directions and intensities of NRM were determined with a spinner magnetometer and used to calculate a core (sample) mean direc-

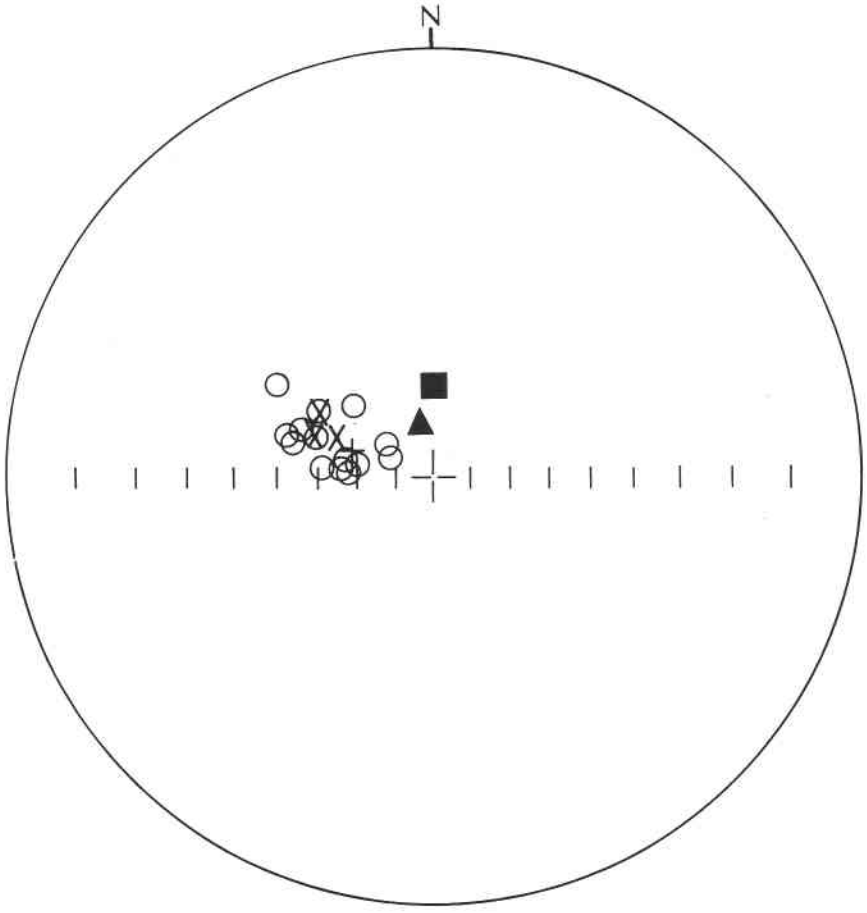


FIG. 2. Stereographic projection showing cleaned sample directions from block samples of the Wilberforce pyroxenite (open circles) together with their mean direction (+). X's are the site-mean directions from the field-drilled cores. North-seeking directions on upper hemisphere. ■ and ▲ are the dipole field direction and the present field direction on the lower hemisphere.

tion. Sample means were combined to calculate site mean directions. All specimens were demagnetized using the newer equipment referred to in the previous paragraph and remeasured after the following peak fields: 50, 100, 200, 300, 400, and 500 Oe. The site mean directions after the 500 Oe field treatment are plotted together with the sample mean directions from the earlier collection of block samples in Fig. 2. Statistical data for the Wilberforce results are recorded in Table 1. Although the early collection of block samples does not constitute a single site, the mean direction from the block samples is treated as a single site in the calculation of the grand mean pole position. This is justified on the basis of the imperfect field orienting procedures used in the collection of the block samples.

Southern Grenville Province—Tudor Gabbro

The geology of the Tudor Gabbro has been described by Lumbers (1969) and the following brief summary is abstracted from his description. The stock is composed mainly of uralitized gabbro and uralitized diorite although other phases including anorthositic gabbro, and partly altered pyroxenite and peridotite are present. Texturally, the plagioclase-bearing

TABLE 1. NRM directions from the Wilberforce Pyroxenite

Site	N	H	R	K	α_{95}	D	I	Pole position
(block samples)		0	No data					
	14	250	13.7712	56.8	5.3°	288.5	−62.0	148.7 E, 18.8 S
1	4	0	3.9951	612.0	3.7°	302.8	−68.4	
	4	500	3.9959	732.0	3.4°	293.6	−62.7	145.2 E, 16.9 S
2	4	0	3.9731	111.0	8.7°	302.3	−60.4	
	4	500	3.9824	170.0	7.1°	299.6	−56.6	146.3 E, 8.6 S
3	5	0	4.9919	496.0	3.4°	301.7	−57.9	
	5	500	4.9932	584.0	3.2°	290.4	−56.9	151.7 E, 13.6 S
							Mean pole	148.0 E, 14.5 S
								$\alpha_{95} = 6.0°$

NOTES: N = number of independently oriented samples per site.
H = demagnetizing field in oersteds.
R = resultant of the N unit vectors (Fisher 1953).
k = best estimate of precision (Fisher 1953).
α_{95} = semi-angle of cone of 95% confidence of mean direction or pole position.
D = declination in degrees east of north.
I = inclination in degrees (positive downward; negative upward).

mafic rocks are generally hypidiomorphic-granular and foliated. The foliation is due to alignment of plagioclase tablets and pyroxene prisms and is a primary flow layering.

No radiometric age dating of the Tudor Gabbro has yet been reported. However, an older limit can be placed on the age of the Tudor Gabbro by virtue of the fact that the Tudor meta-volcanics, into which the gabbro is intruded, contain zircon-bearing rhyolites which yield a uranium–lead age of 1310 ± 15 m.y. (Lumbers 1969, p. 31). Lumbers (1967, 1969) considers that the Tudor Gabbro forms part of a biotite diorite series which is present in the Bancroft–Madoc region. Zircons from the felsic members of this series form a distinct age group at 1250 ± 25 m.y. While this may be the age of emplacement of the Tudor Gabbro, the age of its remanent magnetization which is associated with exsolved magnetite in pyroxene and amphibole (see Stability of NRM and Opaque Petrography) could date from a period of later metamorphism. Other plutons in the area which are interpreted as having been emplaced during the culmination of this late regional metamorphism date at 1125 ± 25 m.y. (Lumbers 1967).

An initial sampling of the Tudor Gabbro was made in 1963 when 12 oriented block samples were taken from road cuts. The NRM directions measured in specimens drilled from these blocks yielded the following result: $N = 12$, $D = 334°$, $I = -13°$, $k = 7$. The mean direction is defined with low precision as was the mean direction determined by Hood (as reported in Cox and Doell (1960)). Alternating field demagnetization was performed on pilot specimens from these block samples using the demagnetization apparatus available in our laboratory in 1963. These experiments removed a substantial present field component and improved the grouping of NRM directions. However, the central area of outcrop of the Tudor Gabbro is characterized by a flow foliation with a rather constant attitude of 320/80° E (Lumbers 1969, Fig. 4) and the cleaned directions were scattered about this plane. Thus the possibility existed that the mean direction obtained was influenced by anisotropy and was not an accurate estimate of the paleofield.

A second field collection was made in 1969 with the samples being field-drilled and oriented with respect to the horizontal and the sun's azimuth. Additional samples were obtained from the localities where the original blocks had been obtained and in addition emphasis was placed on sampling near the margins of the stock where the attitude of the flow foliation was at a large angle to the previously determined NRM direction. Two to 7 independently oriented samples were obtained from each of the 12 sites (Fig. 3). NRM directions and intensities were measured after a.f. demagnetization in the following peak fields: 0, 50, 100, 200, 300, and 400 Oe. Samples from 2 of the 12 sites were poorly grouped initially and

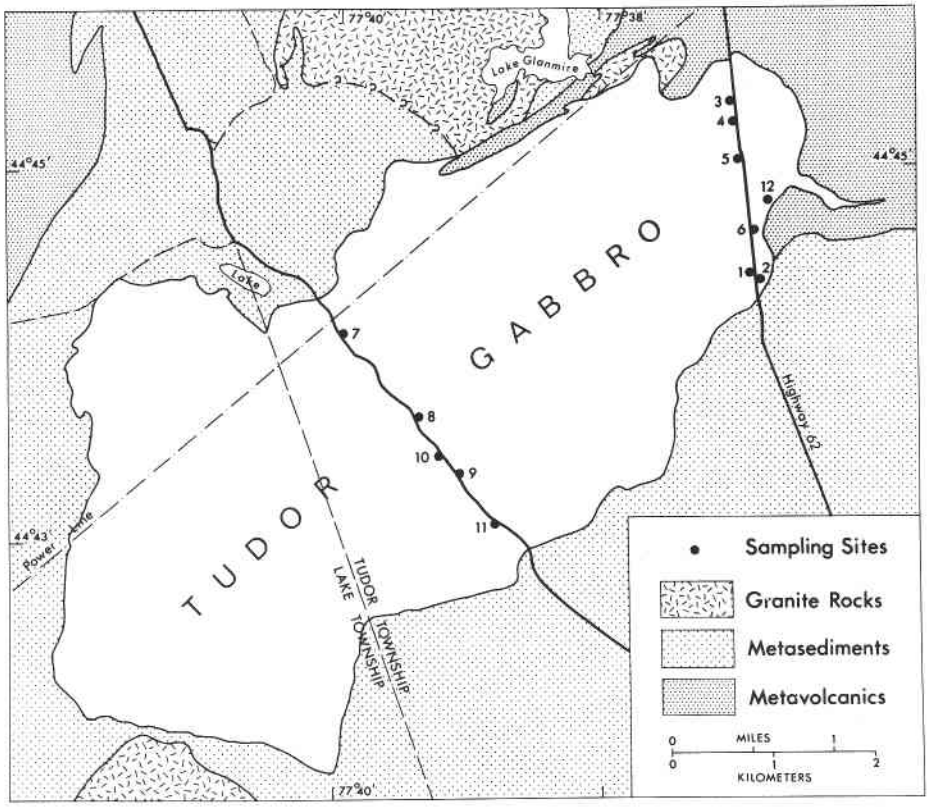


FIG. 3. Sampling sites within the Tudor Gabbro. Geological map simplified after Laakso (1968) and Lumbers (1969).

exhibited widely fluctuating directions and intensities of remanence after partial demagnetization. The samples from the remaining 10 sites, however, proved to be more stable. Within-site directions converged after the 100 or 200 Oe treatment and remained fixed throughout the 300 and 400 Oe treatments. These 10 site mean directions have been plotted in Fig. 4 and the statistical data are recorded in Table 2. For both Fig. 4 and Table 2 the mean direction is that obtained after either the 300 or the 400 Oe treatment, the choice for each site being determined by whichever field produced the greater degree of clustering.

It is to be noted that the overall degree of clustering exhibited in Fig. 4 is not as good as that exhibited by the Wilberforce pyroxenite (Fig. 2) or by the Fall Lake and River Valley anorthosites (Table 3). Since the majority of the Tudor Gabbro samples exhibit a planar foliation it was thought possible that the grains carrying the remanence may have a preferred shape or crystallographic orientation controlled by the planar fabric exhibited by the silicate minerals. For each sample that exhibited a planar fabric or for each site (where the planar fabric was constant in attitude throughout the site) the cleaned remanence direction was plotted together with the plane of foliation. No sympathetic relationship between the remanence directions and planes of foliation was found and thus it is concluded that the accuracy of the result is not adversely affected by the foliation.

Northern Grenville Province—Anorthosite Suite Rocks

The locations of the three rock units which have been sampled from the northern Grenville Province are indicated in Fig. 1.

The meta-anorthosite exposed near the village of St. Charles is a lens-shaped sheet

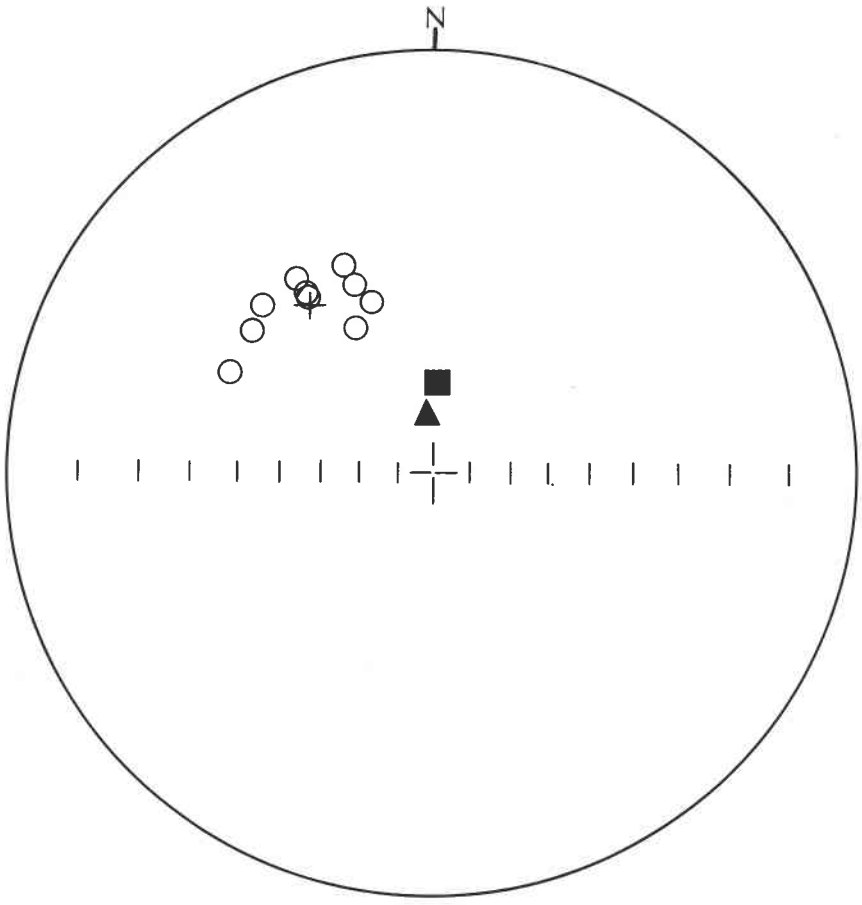


FIG. 4. Stereographic projection showing 10 site-mean directions after magnetic cleaning from the Tudor Gabbro. North-seeking directions are plotted as open circles on the upper hemisphere. + is the mean of the 10 site-means. ■ and ▲ are the dipole field direction and present field direction plotted on the lower hemisphere.

centered on coordinates 80°24′ W, 46°20′ N having an outcrop length of 7 km and a maximum width of 0.6 km (Lumbers 1971*a*, 1972). Although most of the body is intensely foliated, samples were collected from 3 sites from the central portion of the sheet which is massive.

The River Valley anorthosite is the largest anorthositic mass of the three which have been sampled in this study. Geologic mapping in progress (Lumbers 1972, personal communication) indicates that the mass is irregular in plan and has an areal extent of about 190 km². Coarse-grained, massive, dark gray to black gabbroic anorthosite samples were collected from a single site in the body at 80°10′ W, 46°37′ N (Erana Mines quarry site).

Three sites, separated by distances of several hundreds of meters and centered at 79°31′ W, 46°50′ N, were sampled in the Fall Lake body of gabbroic anorthosite. This mass is elliptical in plan with maximum and minimum diameters of 3.2 and 0.8 km (Lumbers 1971*b*,*c*). The long axis of the body trends NE–SW.

Four cores were field-drilled at each of the 7 sites during the summer of 1972. The azimuthal component of orientation for all cores was determined with reference to the sun. The field sampling is not as comprehensive as that

TABLE 2. Mean NRM directions from the Tudor Gabbro

Site	N	H	R	K	α_{95}	D	I	Pole position
1	6	0	4.7038	3.9	39.3°	341.0	+10.6	
	6	300	5.4851	9.7	22.6°	337.2	−38.0	125.1 E, 20.8 N
2	2	0	1.5406	2.2		329.2	+19.8	
	2	300	1.8806	8.4		337.7	−33.1	125.8 E, 24.0 N
3	7	0	6.8059	30.9	11.0°	333.7	−33.3	
	7	300	6.9291	84.6	6.6°	331.9	−46.7	127.7 E, 12.9 N
4	4	0	3.9067	32.1	16.5°	326.5	−28.8	
	4	300	3.9195	37.3	15.2°	324.7	−36.3	137.2 E, 17.5 N
5	5	0	4.4321	7.0	31.0°	336.6	−22.1	
	5	300	4.6069	10.2	25.2°	325.0	−30.6	138.5 E, 21.0 N
6	3	0	2.9524	42.0	19.3°	332.1	−30.5	
	3	400	2.9978	893.0	4.1°	343.7	−43.0	118.0 E, 18.7 N
7	3	0	2.1112	2.3		299.9	+51.0	
	3	300	0.9595	1.0		319.9	+6.0	
8	4	0	3.9508	60.9	11.9°	302.6	−29.4	
	4	300	3.9571	69.9	11.1°	296.1	−33.7	161.0 E, 4.3 N
9	2	0	0.7967	0.8		119.6	+26.6	
	2	300	0.8827	0.9		198.1	+29.2	
10	3	0	2.9983	1163.0	3.6°	309.3	−31.8	
	3	300	2.9968	617.0	5.0°	309.2	−32.2	151.5 E, 12.5 N
11	4	0	3.6782	9.3	31.8°	334.2	−7.7	
	4	300	3.9008	30.3	17.0°	325.7	−35.7	136.4 E, 18.3 N
12	4	0	2.4888	2.0		349.2	+11.4	
	4	400	3.5864	7.3	36.6°	315.4	−30.1	147.2 E, 16.9 N
							Mean pole (N = 10, α_{95} = 8.4°)	137.0 E, 17.1 N

TABLE 3. Mean NRM directions from northern Grenville Province

Site	N	H	R	K	α_{95}	D	I	Pole position
1	4	0	1.2946	1.1	>90°	239.7°	−44.3°	
	4	300	1.5581	1.2	>90°	189.4°	−41.2°	
2	4	0	1.3290	1.1	>90°	300.0°	+49.5°	
	4	300	0.6215	0.9	>90°	357.7°	−50.5°	
3	4	0	3.9479	57.6	12.2°	316.5°	+44.1°	
	4	300	2.1551	1.6	>90°	003.6°	+02.0°	
4	4	0	3.9881	251.0	5.8°	120.7°	+25.4°	
	4	300	3.9849	198.0	6.5°	115.2°	+21.9°	163.6 E, 8.3 N
5	4	0	3.7334	11.3	28.7°	116.0°	+42.7°	
	4	400	3.9883	255.0	5.8°	110.4°	+24.3°	166.9 E, 4.1 N
6	4	0	3.9982	1685.0	2.2°	114.9°	+38.3°	
	4	300	3.9953	639.0	3.6°	122.9°	+27.2°	156.2 E, 10.3 N
7	4	0	3.9843	191.0	6.7°	109.6°	+35.1°	
	4	400	3.9873	236.0	6.0°	120.8°	+25.4°	158.5 E, 9.9 N
							Mean Pole	161.3 E, 8.2 N
								α_{95} = 6.3°

NOTES: Sites 1, 2, 3—St. Charles meta-anorthosite.
Site 4—River Valley anorthosite.
Sites 5, 6, 7—Fall Lake Complex.

done on either of the two southern rock units reported in this study; the results obtained may be considered as of the reconnaissance type. The importance of the results obtained, even if of a reconnaissance nature, stems from the following field observation: the country rocks into which the anorthosites were intruded include rocks which, although much more deformed and recrystallized, are probably equivalent in age to the lower part of the Huronian Supergroup exposed to the north of the Grenville Front (Lumbers 1971*d*, 1972). Thus if

the anorthosites were to possess directions of remanence which yielded 'anomalous' pole positions with respect to the North American curve an interpretation involving large lateral displacements would be suspect on geological grounds.

Each field-drilled core generally yielded 2 specimens plus a chip for petrographic analysis. One polished thin section was made from a core from each site and the sections were studied in transmitted and reflected light. The results of the polished section work are summarized in the next section; a brief résumé of the transmitted light petrography is given below.

Megascopically the St. Charles anorthosite samples are medium-grained, massive, and consist of white feldspar with about 10 to 15% of black ferromagnesian minerals. In thin section the plagioclase is clear, twinned, and weakly zoned. The dark minerals are amphibole and biotite. Accessory minerals include epidote, garnet, and rutile. Opaque oxides are rare; typically only one or two grains are found in any thin section.

The samples from the single River Valley anorthosite site are dark gray, massive rocks with the ferromagnesians only slightly darker than the abundant and coarse-grained plagioclase. In thin section the plagioclase is twinned, inconspicuously zoned, and has a dusty appearance in plane polarized light. Clinopyroxene, invariably with coronas of hornblende, forms 15–20% of the rock. Large opaque oxides are notably absent.

The sampling sites from the Fall Lake body are characterized by medium-grained rocks in which the plagioclase contrasts in color with the dark ferromagnesian minerals. The samples range from anorthositic gabbro to gabbro in composition. In thin section the plagioclase is seen to be clear, twinned, and in some cases, zoned. Clinopyroxenes are unaltered, and in one gabbro section examined, unaltered olivine is present in addition to the pyroxene. All three thin sections examined have abundant Fe–Ti oxide grains, commonly with reaction rims of biotite.

Directions and intensities of remanence were measured in each of the 2 specimens from each core (sample) after a.f. demagnetization in the following peak fields: 0, 50, 100, 150, 200, and 300 Oe. Specimens from the sites in the River Valley anorthosite and the Fall Lake gabbroic anorthosite were remeasured after demagnetization in one further higher field, *i.e.*, 400 Oe peak. Specimen directions were combined to give sample means and the sample means were combined to give site mean directions. The NRM site mean directions and the cleaned site mean directions, together with the standard statistical information, are recorded in Table 3.

Samples from the St. Charles meta-anorthosite are characterized by low remanent intensities and only 3 of the 12 samples have good within-sample consistency of directions. None of the 3 sites exhibits within-site consistency of directions after any of the stages of demagnetization. No significant paleomagnetic result was obtained from the St. Charles meta-anorthosite.

The samples from the River Valley intrusion and the Fall Lake intrusion exhibit a totally different behavior. The virgin NRM directions are generally well grouped initially, remain so or become better grouped with a.f. demagnetization, and the little movement in direction which takes place during demagnetization is very systematic on a within-sample and on a within-site basis. Furthermore, the single site mean direction from the River Valley intrusion is in good agreement with the directions obtained from the Fall Lake sites.

Stability of NRM and Opaque Petrography

The normalized intensities of remanence with alternating field demagnetization for representative samples from the Grenville collection are plotted in Fig. 5. Also plotted for each sample is the value of the stability parameter, S, as defined by Wilson *et al.* (1968). This parameter is a valuable supplement to the J/J_0 vs H curve inasmuch as it takes into consideration directional changes which take place during the demagnetization sequence. In this section an attempt will be made to relate the various stabilities exhibited by the samples to their Fe–Ti oxide petrography.

The Wilberforce pyroxenite samples exhibit high initial intensities of NRM, median destructive fields (m.d.f.) in the 250 to 425 Oe range and values of S vs H only slightly less than J/J_0 vs H. Specimen WS-28-1 is a typical

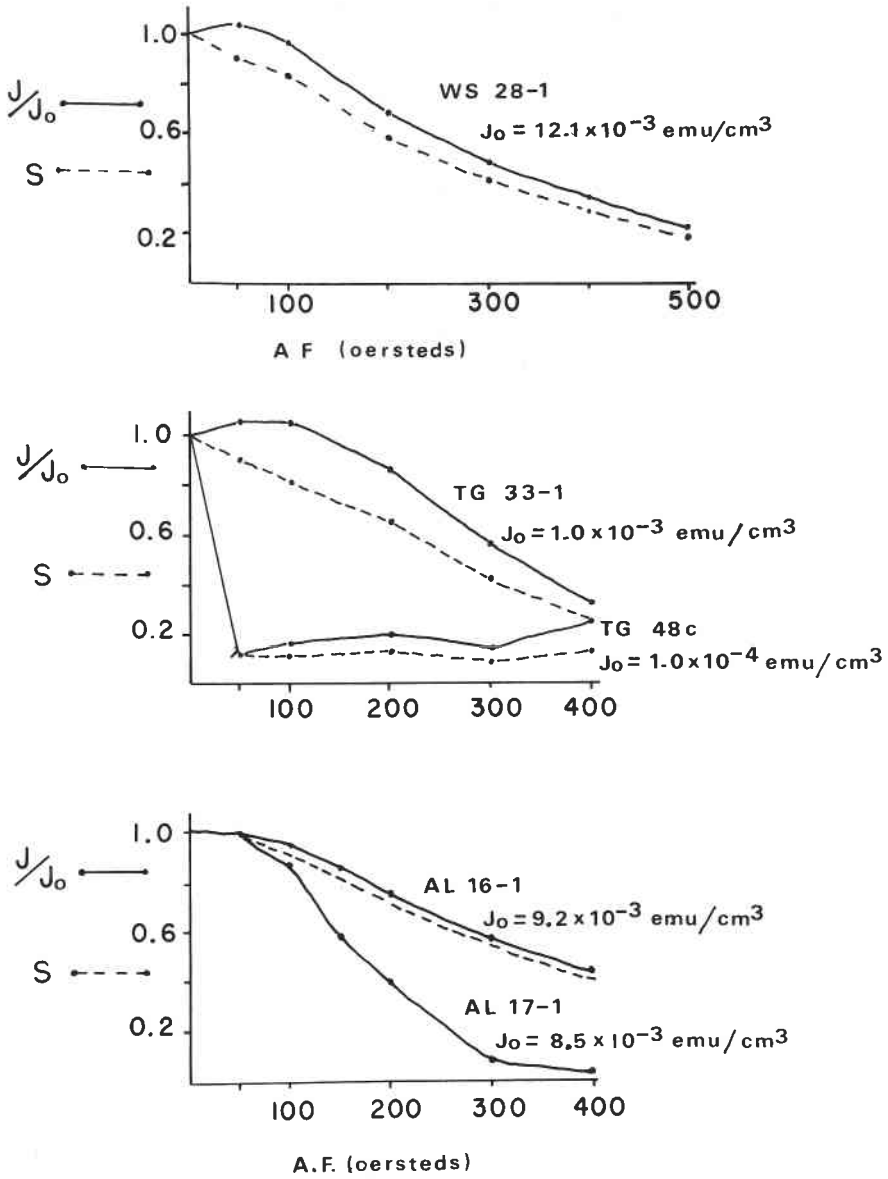


FIG. 5. Normalized a.f. demagnetization curves of NRM drawn as solid curves. *S*, plotted as a dashed curve, is the stability index of Wilson *et al.* (1968). J_0 is the initial intensity of magnetization.

example. The Tudor Gabbro samples exhibit intensities of remanence one or two orders of magnitude less than the Wilberforce samples. Tudor Gabbro samples classed as stable (on the basis of within-site consistency of cleaned directions) are also more variable in their demagnetization characteristics than are those from Wilberforce. Median destructive fields have a broader range (100 to 500 Oe) and many Tudor Gabbro samples exhibit significant differences between J/J_0 and *S*, particularly in the low field ranges. This departure of J/J_0 and *S* (TG-33-1 of Fig. 5 is a typical example) is manifested by a rise in remanence

intensity accompanied by directional changes in the 0–200 Oe range as viscous components are erased.

As seen in polished thin section, the oxides in the Wilberforce samples are of two main types. The first consists of equant grains of homogeneous magnetite having diameters between 20 and 100 μ. The second type, which is volumetrically more abundant, consists of lamellae and rods with axial ratios between 5 to 1 and and 20 to 1 which are included within the pyroxenes and amphiboles. The smallest of these which are optically visible have diameters of about 1 μ. The weight of evidence suggests that the majority of these particles are rod-shaped rather than plate-shaped; when viewed end-on they appear as equidimensional pinpoints of light. The larger of these particles have the optical properties of magnetite. The large accessory opaques in the Tudor Gabbro have considerably more variability than do the large grains in the Wilberforce pyroxenite. Magnetite grains, some with included lamellae of ilmenite (oxidation classes 1, 2, and 3 of Ade-Hall *et al.* (1968), ilmenite grains with and without exsolution lamellae of titanhematite, and pyrite, chalcopyrite, and pyrrhotite are all represented though not typically in the same section. However, in all samples examined from stable sites, the tiny rods of magnetite characteristic of the Wilberforce samples are present. The mode of occurrence is similar although the rods are smaller and not as densely distributed. In fact, they are overlooked at magnifications less than 1000×. The unstable Tudor Gabbro samples (of which TG-48c of Fig. 5 is an example) are hydrothermally altered samples in which the pyroxenes and amphiboles have been totally converted to calcite and chlorite. The only opaque grains in these samples are relatively large, homogeneous magnetite euhedra.

We conclude that the rod-shaped particles in both the Wilberforce pyroxenite and the Tudor Gabbro are the important carriers of the fossil remanence. These particles are similar to the magnetite rods in pyroxenes and amphiboles which Evans and McElhinny (1966) and Evans *et al.* (1968) have described from the Modipe Gabbro. The intensity differences between the Tudor Gabbro and the Wilberforce pyroxenite are qualitatively accounted for by the relative abundance of the amphibole plus pyroxene in these two rock types. Greater variability in the coercive force spectra of the Tudor Gabbro is accounted for by the greater variability in the large opaque grains.

All specimens from the River Valley anorthosite site exhibit practically identical demagnetization curves, of which A1-16-1 is an example. Similarly all specimens from the three Fall Lake sites are similar to each other and A1-17-1 of Fig. 5 is an example. In both of these units the initial NRM directions are well grouped away from the present field direction and the mean direction shifts only slightly with a.f. demagnetization. This feature is illustrated by the coincidence or near coincidence of the J/J_0 vs H curve and the S vs H curve. However, the coercivity spectra are quite different. This difference is manifested in the Fe–Ti oxide petrography. On first inspection one is puzzled as to what carries the very intense and stable remanence in the River Valley anorthosite, inasmuch as large Fe–Ti oxide grains are virtually absent in these rocks. Under high power oil immersion however, rather abundant needle-shaped grains having length–width ratios up to 20/1 are seen. The smallest which are visible under this magnification have diameters <1 μ. The mode of occurrence and optical properties are identical to those described by Murthy *et al.* (1971) in the Michikamau anorthosite. The Fall Lake anorthosite gabbro samples, in contrast, have abundant large (1 mm diameter) opaque grains. These large grains are subdivided however on two size scales. The large dividers are broad lamellae of ilmenite. The host is not homogeneous but is subdivided in the cloth textured pattern of magnetite and ulvospinel. It is this latter textural subdivision which is believed responsible for providing the moderate stability of NRM. Note that most of the remanence in the Fall Lake samples is destroyed in the 100 to 300 Oe range but that relatively little is lost in the 0 to 100 Oe range. Although the coercivity spectra of these two anorthositic bodies are quite different as are the Fe–Ti oxide assemblages it is noteworthy that the remanence directions and the pole positions derived from these directions are essentially identical.

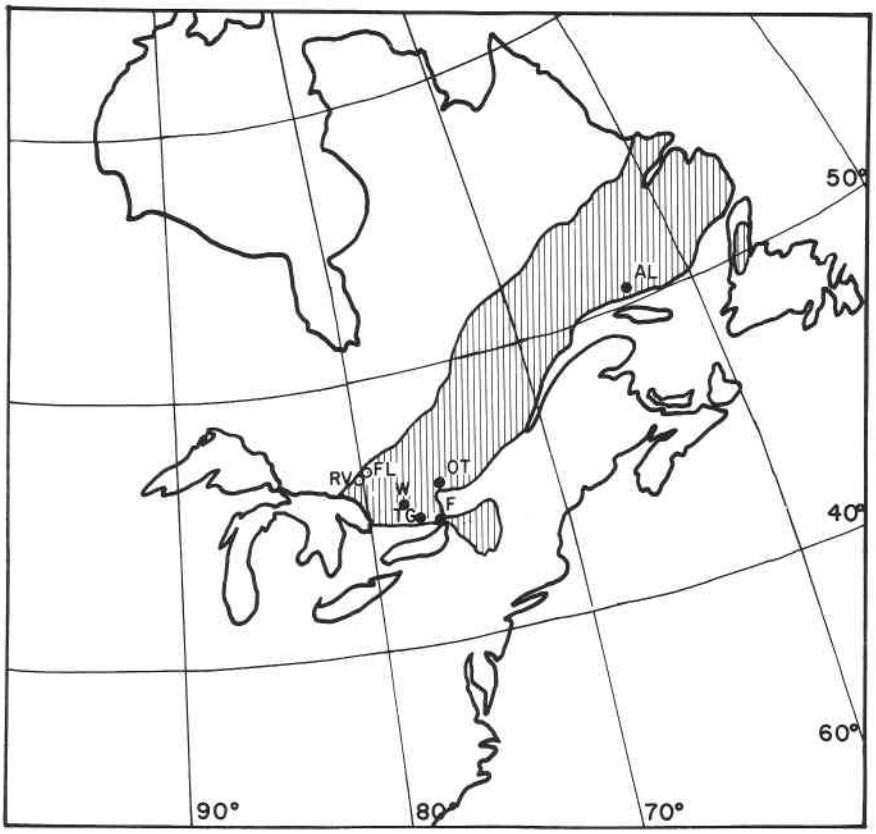


FIG. 6. The Grenville Province of eastern North America is outlined by vertical shading. Dots and circles indicate the location of rock units which have yielded paleomagnetic results: AL, the Allard Lake anorthosite (Hargraves and Burt 1967); OT, Ottawa area basic intrusives (Irving *et al.* 1972); F, Frontenac axis dikes (Park and Irving 1972); TG, Tudor Gabbro (this paper); W, Wilberforce pyroxenite (this paper); FL, Fall Lake anorthosite gabbro; RV, River Valley anorthosite (this paper).

Paleomagnetic Poles and Possible Tectonic Implications

Two recent estimates (Spall 1971; Robertson and Fahrig 1971) of the polar wander path with respect to North America for late Precambrian time are reproduced in Fig. 7. Both curves indicate little apparent change in the field configuration over two fairly long time intervals: 1475–1200 m.y. and 1050–700 m.y. Rapid polar movement is indicated for the relatively short interval: 1200–1100 m.y. However, the magnitude of this polar movement is not yet resolved and this is shown by the departure of the two curves in this time interval. The departure of the curves in this time range arises from different criteria of data selection, different interpretation of the published data, and differing access to unpublished results from two main bodies of data.

One of these bodies of data is the paleomagnetic result from the Abitibi dikes (Larochelle 1966) the age of which was initially reported as 1230 m.y. on the basis of two K–Ar determinations from biotite from one of these dikes. At the time of Spall's analysis the Abitibi result appeared anomalous when compared to other North American results of this approximate age and the Abitibi pole was not incorporated into the trend of the curve. Robertson and Fahrig (1971) cited reasons for revising the age downward to about 1150 m.y. and incorporated the result into the trend

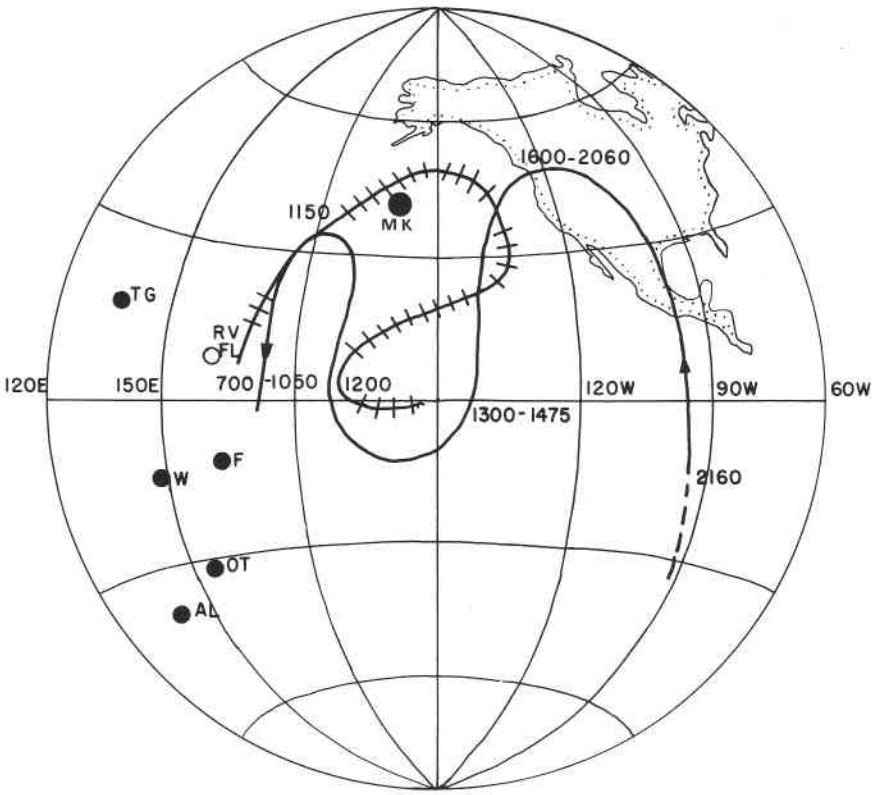


FIG. 7. Equal area projection showing two recent estimates of the North American paleomagnetic polar curve for portions of Precambrian time. The solid curve is after Spall (1971) and covers the time interval 700–2160 m.y. The curve with cross bars covers the shorter time interval 700–1500 m.y. and is after Robertson and Fahrig (1971). Dots and circles are the pole positions for the Grenville units shown in Fig. 6. Pole MK is that given by Palmer (1970) for the Middle Keweenawan lavas and sills of the Lake Superior region.

of their curve. Subsequently Rb/Sr isochron studies yielded an age of 2147 ± 68 m.y. for these dikes (Gates 1971, 1972). This age is very close to that of the Nipissing diabase yet the Nipissing diabase and the Abitibi dikes have very different directions of magnetization. Thus it is still not clear whether the age of the magnetization of the Abitibi dikes should be keyed to the Rb/Sr 'age', the K–Ar 'age' or to neither.

The second body of data which results in the departure of the two curves concerns the interpretation of the large amount of paleomagnetic data from the Middle Keweenawan of the Lake Superior region which basically yields two sets of pole positions associated with normally magnetized and reversely magnetized lava flows and mafic intrusions. Stratigraphic evidence indicates that the reversely magnetized units are older and they yield rather scattered poles in the northeastern Pacific. The normally magnetized units yield more tightly grouped pole positions in the north-central Pacific. The curve of Robertson and Fahrig (1971) incorporates both sets of poles whereas that of Spall incorporates only the normally magnetized results. Palmer (1970) favored the interpretation that the non-coincidence of the normal and reversed poles was the result of a non-removable secondary component of magnetization which was revealed by a conglomerate test. On this assumption an average of the normal and reversed poles (unit weight given to each) was considered to give the best estimate for the pole position in Middle

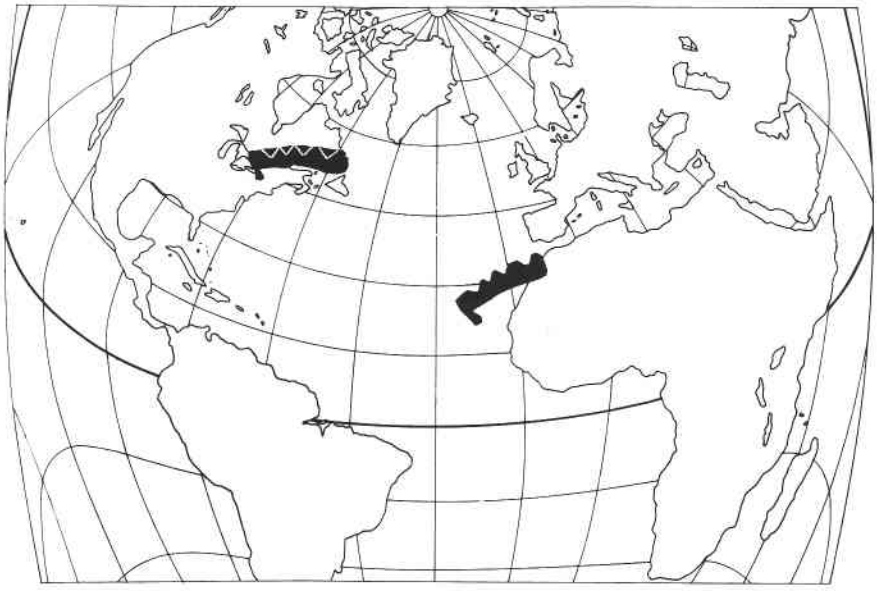


FIG. 8. A suggested former position of the southern Grenville Province (outlined in black and truncated to the north with saw teeth) has been obtained by shifting poles AL, OT, W, and F of Fig. 7 to the 1200–1475 portion of the North American curve and shifting TG to MK of Fig. 7. The southern Grenville Province block is shifted accordingly.

Keweenawan time (1120–1140 m.y. (Silver and Green 1972)) and this pole is plotted as MK in Fig. 7.

The striking feature of Fig. 7 is that the Grenville poles plot well to the west of the North American curves, with the exception of the one from the River Valley and Fall Lake bodies that are situated in the northern part of the province. The locations of the rock units whose poles are plotted in Fig. 7 are shown in Fig. 6. There is some geological evidence that the Northern Grenville Province bodies are intruded into rocks which may be correlative with rocks to the north of the Grenville Front. However, no similar geological relationships are indicated for the southern rock units which yield the 'anomalous' poles. Following this line of reasoning we suggest that the magnetization of these southern units was acquired in the Late Precambrian (probably prior to the 950 m.y. K–Ar Grenville event) but when it was acquired the southern Grenville Province was situated several thousand kilometers from its present position.

In an effort to obtain an estimate of its former position, the polar wander curves, together with the raw data used to construct these curves, were transferred to plastic hemispheres moulded to fit a 16 in. (40.6 cm) globe. The anomalous Grenville poles were plotted on a second hemispherical overlay. The most convincing match of the data was obtained by fitting the 'cluster' formed by poles AL, OT, W, and F to the North American cluster of poles situated in the central equatorial Pacific (1200–1475 m.y. interval) and the lone Tudor Gabbro pole to the 1120–1140 m.y. Middle Keweenawan pole (MK of Fig. 7). This matching of poles places the southern Grenville Province in the position indicated in Fig. 8. This position is not radically different from that obtained by Irving *et al.* (1972) by matching AL, OT, and F to the descending limb of Robertson and Fahrig's curve. Many of the same conclusions follow: components of convergent and dextral strike–slip motion are required to bring the two blocks into juxtaposition; the two blocks were separate prior to 1150 m.y. and the K–Ar dates which peak at 950 m.y. correspond to a period of general uplift subsequent to the collision of the two blocks; the suture between the two blocks lies

south of the Labrador iron formations in Quebec and south of the Fall Lake and River Valley anorthosites in Ontario.

Whether this highly speculative interpretation stands the test of future investigations remains to be seen. Geochronological data utilizing K–Ar, Rb/Sr, and Pb/Pb systems are sorely needed. Additional paleomagnetic data from other regions of North America, particularly from the 1050–900 m.y. time interval would help rule out the possibility that the 'anomalous' Grenville poles simply represent a brief excursion of the field which has as yet been unrecorded elsewhere. Failing this possibility the paleomagnetism of the Grenville Province provides persuasive evidence for late Proterozoic continental drift.

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